HCMC UNIVERSITY OF TECHNOLOGY AND EDUCATION
Faculty for High Quality Training

Program name: Automation and Control Engineering Technology

FINAL EXAMINATION
SEMESTER 1 – ACADEMIC YEAR 2020-2021
Course name: Applied Mathematical in Electrical Engineering

Course ID: AMEE331944E
Exam code: 01 Number of pages: 02
Duration: 90 minutes.
Open-book.

Question 1 (2.5 marks): Given the linear circuit in Figure 1, show the details of your work in

a) modelling the linear circuit by three equations in three unknown branch currents(i_1, i_2, i_3) (Use Kirchhoff's law),

b) determining the determinant of the coefficient matrix $\boldsymbol{A}$ and the inverse of the matrix $\boldsymbol{A}$ by determinants and

c) finding i_1 by using Gauss elimination.

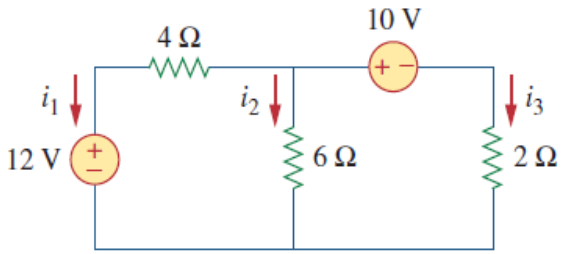

Figure 1

Question 2 (2.5 marks): Given the linear AC circuit in Figure 2 with $R = 100\Omega, C = 10^{-6}F$ and $i_s(t) = 0.5\cos 10^3 t\ (A)$, show the details of your work in:

a) modelling the linear ac circuit by two linear ordinary differential equations when i and v are unknown (Use Kirchhoff's law),

b) determining the two equivalent equations by using complex numbers and

c) finding the complex current I by using Cramer's rule and $i(t)$.

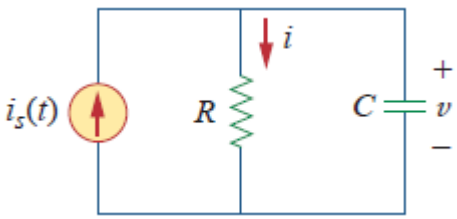

Figure 2

Question 3 (2.5 marks): Given the circuit in Figure 3, where $u(t)$ is the unit step function, $i(0) = 0, v(0) = 0$. Show the details of your work in

a) modelling the linear AC circuit by two linear ordinary differential equations when i and v are unknown,

b) determining the equivalent equations by using Laplace transform and

c) finding I by using Cramer's rule and $i(t)$ by using inverse Laplace transform.

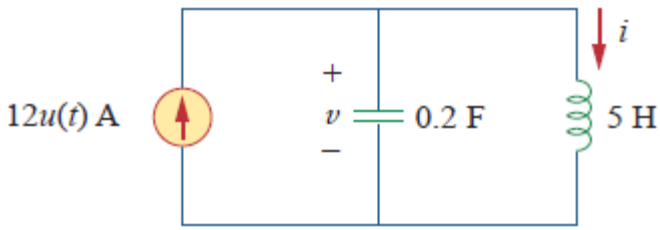

Figure 3

Question 4 (2.5 marks): Given the periodic function $f(t)$ in Figure 4, show the details of your work in:

a) finding the trigonometric Fourier series coefficient a_n and b_n of the function shown in figure 4,

b) calculating the amplitude A_n and phase φ_n and

c) plotting the amplitude and phase spectra.

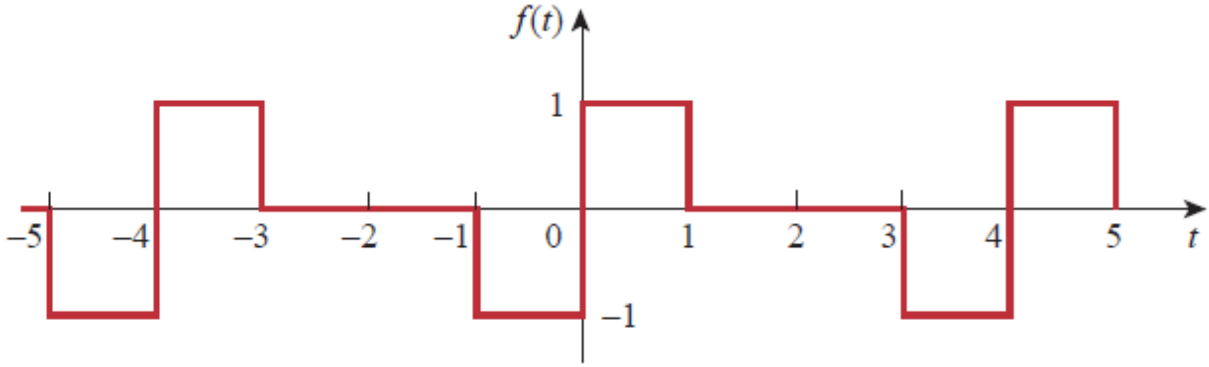

Figure 4

Note: Proctors are not allowed to give any unauthorized explanations.

Learning outcome mapping	**Assessed in**
[LO 1.1]: Modeling of electrical engineering problems by linear equations, matrix theories.	Questions 1
[LO 2.1]: Modeling of electrical engineering problems by differential equations, matrix theories.	Questions 2, 3
[LO 2.4]: Solving matrix-related problems	Question 1,2, 3
[LO 2.5]: Performing analyses with complex numbers	Question 2
[LO 2.3]: Performing analyses with Laplace transform	Question 3
[LO 2.6]: Analysis and argumentation to solve problems by Fourier series	Question 4

18/01/2021

Approved by Program Chair

(signed and named)